

Zihao Liu

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EDUCATION

University of Cambridge

Cambridge, UK

*BA (Hons) and MEng in Engineering (Year 2); **Current Grade: 2:1 (projected First Class)** Oct 2024 – Sept 2028*

Core modules: Mechanics; Structures; Materials; Thermofluid Mechanics; Electrical Engineering; Information Engineering; Mathematical Methods

Year 2 Relevant coursework: Integrated Design Project (robot vehicle design and programming); Earthquake-Resistant Structures (lab experiments, risk assessment, design project); Sustainable Engineering Poster Project; Mechanics and Thermofluid Experiments; Electrical and Information Engineering Labs; Structures and Materials Experiments; Computing and Data Science Labs (device programming, time-series analysis).

Year 1 Foundation: Integrated Electrical Project (oscilloscope, analogue/digital circuits, LTSpice, PicoScope); Product Design Project (full design cycle from concept to prototype); Structural Design Project (steel truss analysis and construction); Engineering Drawing (orthographic/isometric); SolidWorks Motion Simulation Project.

National University of Singapore

Singapore

*EE111B – Electrical & Computer Engineering Principles and Practice; **Grades: A***

May 2022 – Jul 2022

- Completed a course in electrical and computer engineering, gaining hands-on experience in electronic circuits, computer-aided systems, and robotics. Built a sensor-assisted autonomous robotic vehicle as a final project.

Hwa Chong Institution

Singapore

Singapore-Cambridge GCE A Level

Jan 2020 – Dec 2023

- Grades: 4A (Maths, Physics, Chemistry, Economics); Distinction in NTU-H3 Semiconductor Physics and Devices

SKILLS

Languages: C/C++, Java, Python, JavaScript, TypeScript, Dart, Kotlin, SQL

Software Technologies: Git, Linux, Angular, Flutter, Node.js, Flask, Figma, Spring Boot, AWS, REST APIs, Python libraries (NumPy, Pandas, Matplotlib, SciPy), **PyTorch**, **TensorFlow**, Scikit-learn, SQL

Engineering Technologies: Arduino, SolidWorks, KiCad, LTSpice, PicoScope

Methodologies: Agile, Scrum, OOP, MVC Framework and Design Patterns, Functional Programming, DevOps, CI/CD

EXPERIENCE

Minimax Labs Ltd

London, UK

Software Engineer

Jun 2025 – Sept 2025, Full-time Internship

- Contributed to the development of LiNGO, a large-scale LNG portfolio and logistics optimisation platform and DataHub, a database system that stores and manages the large number of data required for the optimisation algorithms, by implementing new frontend features and improving system reliability in a professional software engineering environment.
- Designed and built UI components using **TypeScript**, **Angular**, and **RESTful APIs**, with backend integration in **Java**.
- Enhanced usability and maintainability through reusable components, automated saving mechanisms, and system update features.
- Applied **Git** workflows and collaborated with senior engineers in both in-person and remote settings.
- Strengthened knowledge of **software architecture**, **OOP design patterns**, dependency management, and UI/UX design principles.

Cambridge University Chinese Undergraduate Society (CCS)

Cambridge, UK

DevOps Engineer; IT Department Coordinator

Oct 2024 – Present, Part-time

- Designed and implemented core features for the Society's **WeChat Mini Program** using **JavaScript**, integrating interactive functions to enhance user engagement.
- Maintained and optimised the backend database, ensuring data integrity, reliability, and scalability for hundreds of active users.
- Monitored and tracked system performance, detected anomalies, and supported timely incident response.

- Led continuous deployment and feature improvement cycles, applying **DevOps** best practices to shorten development feedback loops.
- Collected and analysed user feedback to prioritise enhancements and improve user experience.
- Provided technical support and solutions for public events, including device setup, live troubleshooting, and staff coordination.

Full Blue Racing

Cambridge, UK

EV Battery Engineer

Oct 2024 – Present, Part-time

- Designed and developed the battery system for the electric racing car, including battery cells, **Battery Management System (BMS)**, Tractive System Accumulator Container (TSAC), and charging system, ensuring high reliability and performance for Formula SAE Electric competition.
- Analysed design requirements and system specifications to define the functional and safety needs of the BMS, including cell balancing, state-of-charge monitoring, and fault detection.
- Prototyped and designed BMS circuits using KiCad, iterating on schematics and layouts to meet electrical and mechanical constraints. Collaborating with the EV team to integrate the battery system into the vehicle.

Nanyang Technological University

Singapore

Student Researcher

Apr 2022 – March 2023, Internship

- Awarded **Gold Award** for the project “*Data-Driven Method for Li-Ion Battery Health Monitoring*”, mentored by [Prof. Yan Xu](#), School of Electrical and Electronic Engineering, NTU.
- Designed and implemented a ML framework using **LSTM** and **GRU** models to predict battery State-of-Health (SOH), addressing limitations of previous methods (high complexity, low interpretability, high training cost).
- Developed and validated the model in **PyTorch**, achieving high accuracy with an average **RMSE of 0.724%** on battery cell datasets. This project is open-sourced on [GitHub](#).

Hwa Chong Infocomm and Robotics Society

Singapore

Machine Learning Section Leader

Feb 2022 – Aug 2023

- Provided training to society members on deep learning with **TensorFlow**, focusing on **time series** analysis and **NLP** applications.
- Organised academic workshops and outreach programmes to promote AI and robotics.
- Maintained and managed Linux servers for the society’s projects and members.

AWARDS & ACHIEVEMENTS

Cambridge Trust Scholarship: Prestigious merit-based award supporting outstanding international students with their academics at the University of Cambridge. (Oct 2024 - Sept 2028)

Nanyang Research Programme (NRP) 2022 – Gold Prize: Awarded for the project “*Data-Driven Method for Li-Ion Battery Health Monitoring*” at NTU, Singapore. (Apr 2022 – Mar 2023)

PROJECTS

Awesome Portfolio Page | [GitHub](#)

- A **React**-based web application for creating a fully customizable portfolio site to showcase professional experience, projects, and contact information in a clean, responsive layout.
- Implemented structured data-driven content management through a single JSON configuration file, ensuring seamless updates and validation. Optimised for deployment on GitHub Pages, Netlify, and Vercel.

Intelligent System to Prevent Contamination of Recyclables

Jun 2023 – Sept 2023

- Developed as part of the Sustain-a-City: An Urban Innovation Challenge, achieving **2nd runner-up**, by designing an integrated recycling system to reduce waste contamination.
- Built a **YOLOv8**-powered smart dustbin prototype on **Raspberry Pi**, enabling accurate recyclable vs. non-recyclable classification.

High School Mathematical Contest in Modeling (HiMCM) 2022

Oct 2022 – Nov 2022

- Modelled the global relationship between temperature and CO₂ using regression, time-series, and grey model analysis. Implemented modelling in **PyTorch** and statistically compared effectiveness.

DSTA Today-I-Learned AI Competition (2nd Runner-up) | [GitHub](#)

May 2022 – Jun 2022

- Developed and trained **computer vision (CV)** and **natural language processing (NLP)** models in **PyTorch**, and deployed them on a robot to perform tasks such as object detection and voice recognition with robust accuracy.

REFeree

Dr. Simone Hochgreb, Director of Studies, Department of Engineering, University of Cambridge; [Email](#)